Kruskal's Algorithm for Communication Tower Network

Minimum Spanning Tree (MST) using C++

1. Problem Statement

Develop a C++ program to represent communication towers as vertices and candidate communication links as weighted edges in a graph. Apply Kruskal's algorithm to connect all communication towers at minimum total cost by constructing a Minimum Spanning Tree (MST). Display the selected links, their weights, the total cost, and the average weight of the selected links.

2. Objective

1. To represent a communication network as a weighted graph, with towers as vertices and links as edges.
2. To apply Kruskal's algorithm (greedy approach) with cycle detection using the Union-Find method.
3. To connect all towers at minimum total cost using exactly V - 1 links.
4. To calculate and display the total cost and the average weight of the selected links.

3. Algorithm

Step 1: Start the program.

Step 2: Input the number of communication towers (vertices) and candidate links (edges).

Step 3: Assign a weight to each link representing its installation cost or distance.

Step 4: Sort all edges in ascending order of their weights.

Step 5: Initialize the Minimum Spanning Tree (MST) as empty and set the total cost to zero.

Step 6: Select the edge with the minimum weight from the sorted list.

Step 7: Check whether adding the selected edge creates a cycle. If no cycle is formed, add the edge to the MST and update the total cost. If a cycle is formed, discard the edge.

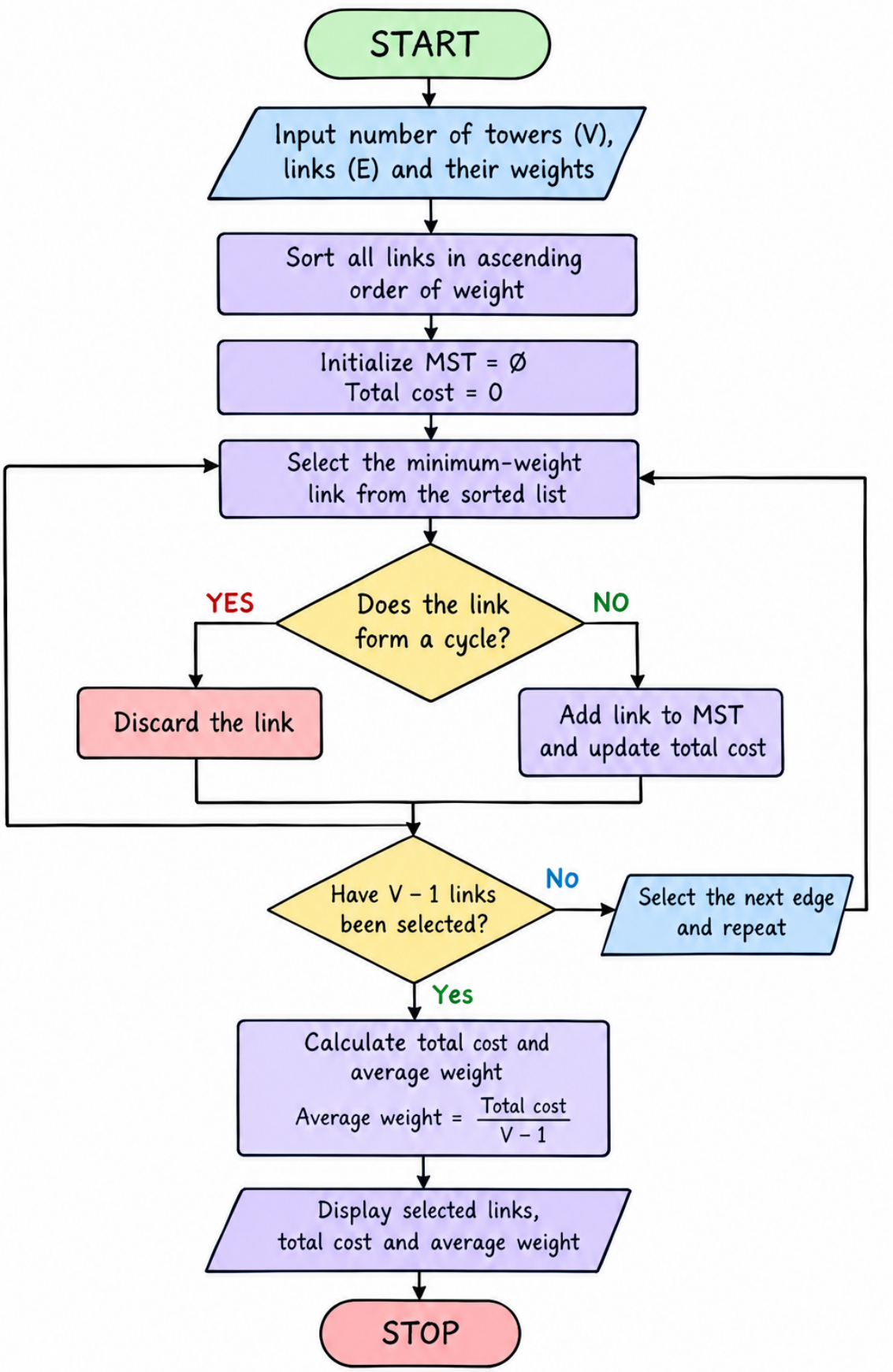
Step 8: Repeat Steps 6 and 7 until V - 1 edges have been selected, where V is the number of towers.

Step 9: Display the selected links and their weights.

Step 10: Calculate and display the total cost and average weight of the selected links (Average weight = Total cost / (V - 1)).

Step 11: Stop the program.

4. Flowchart


START
Input number of towers (V), links (E) and their weights
Sort all links in ascending order of weight
Initialize MST = Ø
Total cost = 0
Select the minimum-weight link from the sorted list
Does the link form a cycle?
YES
NO
Discard the link
Add link to MST and update total cost
Have V – 1 links been selected?
No
Select the next edge and repeat
Yes
Calculate total cost and average weight
Average weight = Total cost / V – 1
Display selected links, total cost and average weight
STOP